CIE AS PHYSICS (9702) • QUESTION PAPER

5.1 Energy Conservation

Syllabus Section: 5. Work, Energy & Power **Paper Type:** Multiple Choice (Paper 1) **Total:** 34 Questions (34 Marks)

Question 1 Easy [1 mark]

Which of the following is a statement of the principle of conservation of energy?

A energy is the product of force and distance

B energy cannot be created, destroyed or transformed

C the total energy of an isolated system is constant

D the power transmitted by a system is proportional to the rate of transfer of energy

Question 2 Easy [1 mark]

Which expression **defines** power?

A force × displacement in the direction of the force

B force × velocity

C work done ÷ time taken

D work done × time taken

Question 3 Easy [1 mark]

The total energy input E_{in} in a process is partly transferred to useful energy output U, and partly to energy that is wasted W

What is the efficiency of the process?

A $\frac{U}{W} \times 100\%$

B $\frac{W}{E_{in}} \times 100\%$

C $\frac{U}{E_{in}} \times 100\%$

D $\frac{U+W}{E_{in}} \times 100\%$

Question 4 Easy [1 mark]

A boat moving at constant speed v through still water experiences a total frictional drag F

What is the power developed by the boat?

A $\frac{1}{2}Fv$

B Fv

C $\frac{1}{2}Fv^2$

D Fv^2

Question 5 Easy [1 mark]

What is the average power output of a laser that can deliver 0.20 J of energy in 10 ns?

A 2 nW

B 20 mW

C 200 kW

D 20 MW

Question 6 Easy [1 mark]

A transformer has the following input and output

	Potential difference / V	Current / A
Input	11 000	28
Output	240	1200

What is the efficiency of the transformer?

A 0.94 %

B 1.0 %

C 11 %

D 94 %

Question 7 Easy [1 mark]

Which operation involves the greatest mean power?

A a car moving against a resistive force of 0.4 kN at a constant speed of 20 m s^{-1}

B a crane lifting a weight of 3 kN at a speed of 2 m s^{-1}

C a crane lifting a weight of 5 kN at a speed of 1 m s^{-1}

D a weight being pulled across a horizontal surface at a speed of 6 m s^{-1} against a frictional force of 1.5 kN

Question 8 Easy [1 mark]

An electric motor is required to produce 120 W of mechanical output power. The efficiency of the motor is 80%

Which row is correct?

	Electrical power input to motor / W	Waste heat output from motor / W
A	120	24
B	120	96
C	150	30
D	150	120

A

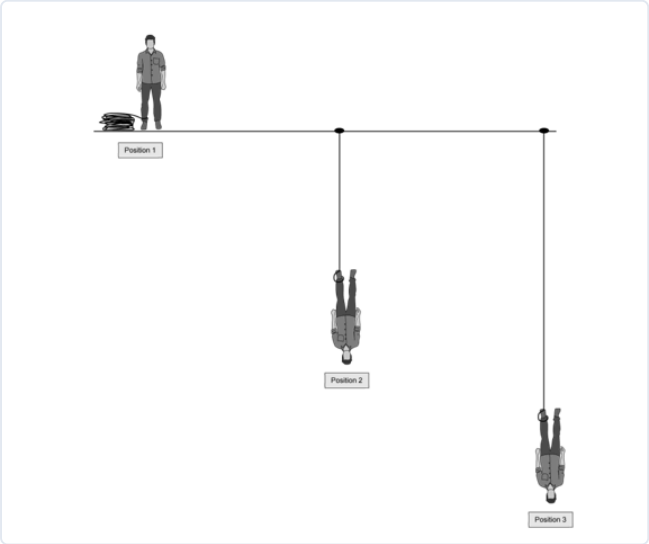
B

C

D

Question 9 Easy [1 mark]

When bungee jumping, a student starts with maximum gravitational potential energy (position 1), then falls freely until the rope fully unwinds (position 2), after which the rope starts to stretch until the lowest point of the jump is reached (position 3)



What are the kinetic and elastic potential energies at position 3?

	Kinetic energy	Elastic potential energy
A	maximum	maximum
B	maximum	minimum
C	minimum	maximum
D	minimum	minimum

A

B

C

D

Question 10 Easy [1 mark]

The kinetic energy of a moving vehicle with mass 1000 kg is 4.5×10^5 J. It brakes with a total constant braking force of 6000 N

What will be its stopping distance?

A 37 m

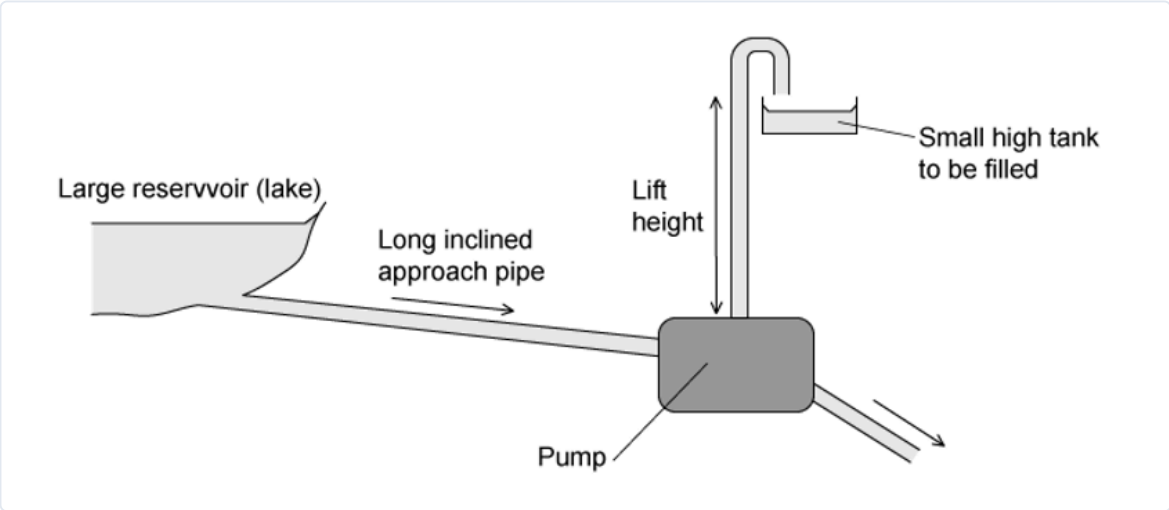
B 75 m

C 150 m

D 300 m

Question 11 Medium [1 mark]

The diagram shows a pump called a hydraulic ram



In one such pump the long approach pipe holds 500 kg of water. A valve shuts when the speed of this water reaches 2.0 m s^{-1} and the kinetic energy of this water is used to lift a small quantity of water by a height of 15 m.

The efficiency of the pump is 10 %.

Which mass of water could be lifted 15 m?

A 0.15 kg

B 0.68 kg

C 1.5 kg

D 6.8 kg

Question 12 Medium [1 mark]

Trains supply coal to a power station. The table below gives quantities describing the operation of the power station.

	Symbol	Unit
power station output	*P*	W
number of trains per day	*N*	
mass of coal on a train	*M*	kg
energy from 1 kg of coal	*J*	J
number of seconds in one day	*S*	s

Which expression gives efficiency of the power station?

A $\frac{PS}{NMJ}$

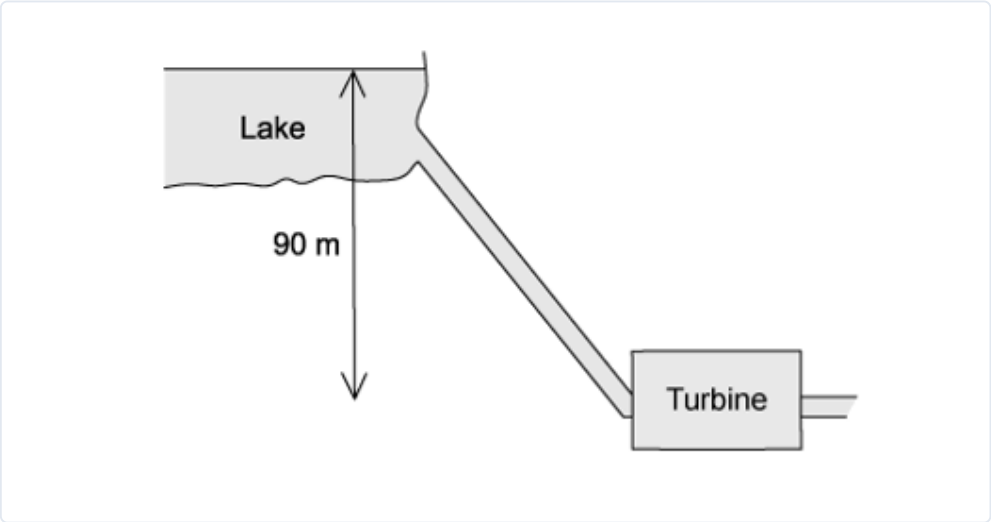
B $\frac{PSN}{MJ}$

C $\frac{NMJ}{PS}$

D $\frac{NM}{PSJ}$

Question 13 Medium [1 mark]

Water flows from a lake into a turbine that is a vertical distance of 90 m below the lake, as shown



The mass flow rate of the water is 2400 kg min^{-1}. The turbine has an efficiency of 75 %.

What is the output power of the turbine?

A 26 kW

B 35 kW

C 1.6 MW

D 2.1 MW

Question 14 Medium [1 mark]

Initially, four identical uniform blocks, each of mass *m* and thickness *h*, are spread on a table.



How much work is done on the blocks in stacking them on top of one another?

A 3 mgh

B 6 mgh

C 8 mgh

D 10 mgh

Question 15 Medium [1 mark]

The first column in the table gives four examples of work being done. The second column gives more detail of the action.

Which row is **not** correct?

	Example	Detail
A	a girl dives from a diving board into a swimming pool	work is done by the girl against gravity as she falls
B	a man pushes a car along a level road	work is done by the man against friction
C	an electron is accelerated towards a positively-charged plate	work is done on the electron by the electric field of the plate
D	a piston is pushed outwards as a gas expands	work is done on the atmosphere by the gas

A

B

C

D

Question 16 Medium [1 mark]

The data below are taken from a test of a petrol engine for a motor car.

power output	150 kW
fuel consumption	20 litres per hour
energy content of fuel	40 MJ per litre

Which expression will evaluate the efficiency of the engine?

A $\frac{150 \times 10^3}{40 \times 10^6 \times 20 \times 60 \times 60}$

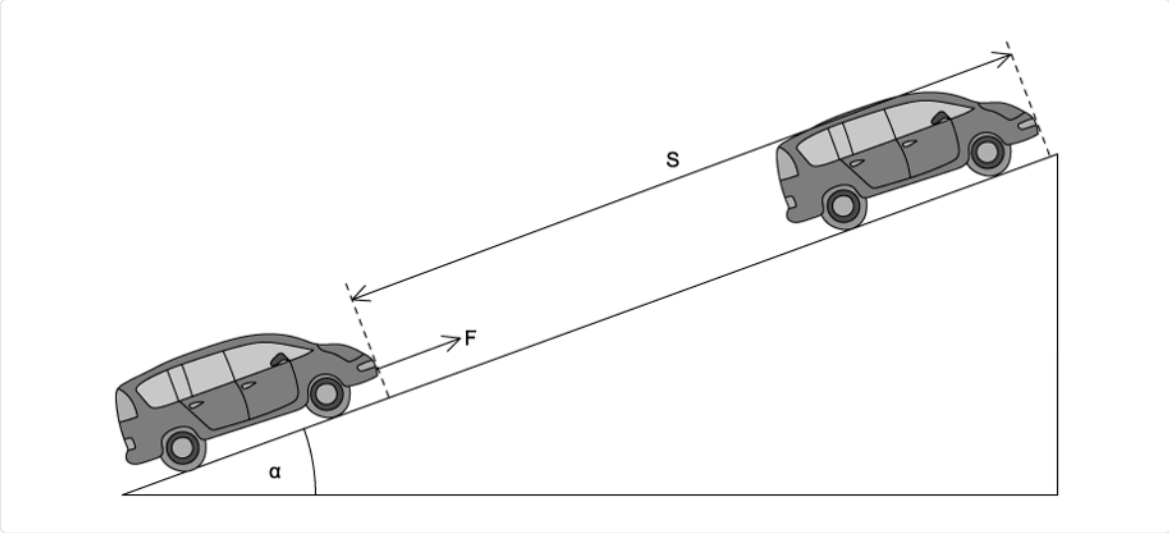
B $\frac{150 \times 10^3 \times 60 \times 60}{20 \times 40 \times 10^6}$

C $\frac{150 \times 10^{3 \times} 40 \times 10^6 \times 20}{60 \times 60}$

D $\frac{150 \times 10^3 \times 20}{40 \times 10^6 \times 60 \times 60}$

Question 17 Medium [1 mark]

A constant force F acting on a car of mass m moves it up a slope through a distance s at constant velocity v. The angle of the slope to the horizontal is α.



Which expression gives the efficiency of the process?

A $\frac{mgs \sin \alpha}{Fv}$

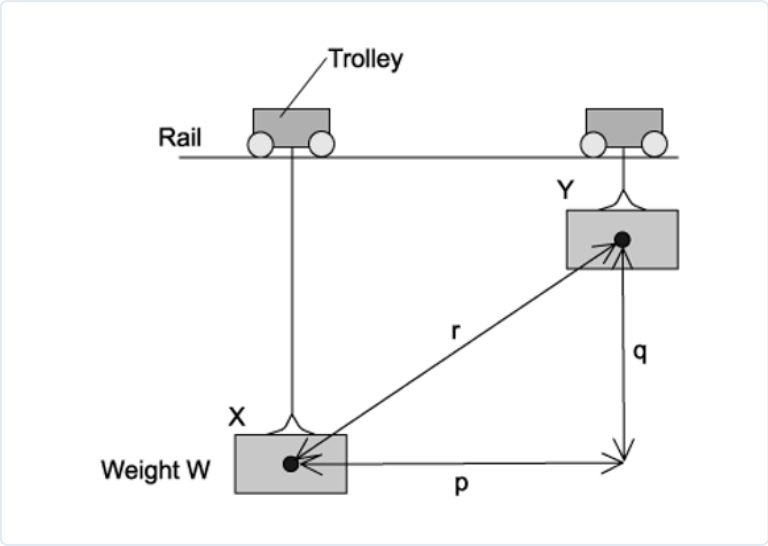
B $\frac{mv}{Fs}$

C $\frac{mv^2}{2Fs}$

D $\frac{mg \sin \alpha}{F}$

Question 18 Medium [1 mark]

A weight *W* hangs from a trolley that runs along a rail. The trolley moves horizontally through a distance *p* and simultaneously raises the weight through a height *q*.



As a result, the weight moves through a distance *r* from X to Y. It starts and finishes at rest.

How much work is done on the weight during this process?

A *Wp*

B *W(p + q)*

C *Wq*

D *Wr*

Question 19 Medium [1 mark]

A bungee jumper has 24 kJ of gravitational potential energy at the top of his jump. He is attached to an elastic rope which starts to stretch after a short time of free fall. Assume that energy loss through air resistance is negligible.

	GPE / kJ	EPE / kJ	KE / kJ
Top	24	0	0
Bottom	0	24	0

What are the possible values when the jumper is half-way down?

	GPE / kJ	EPE / kJ	KE / kJ
A	12	10	2
B	12	8	4
C	8	8	8
D	12	2	10

A

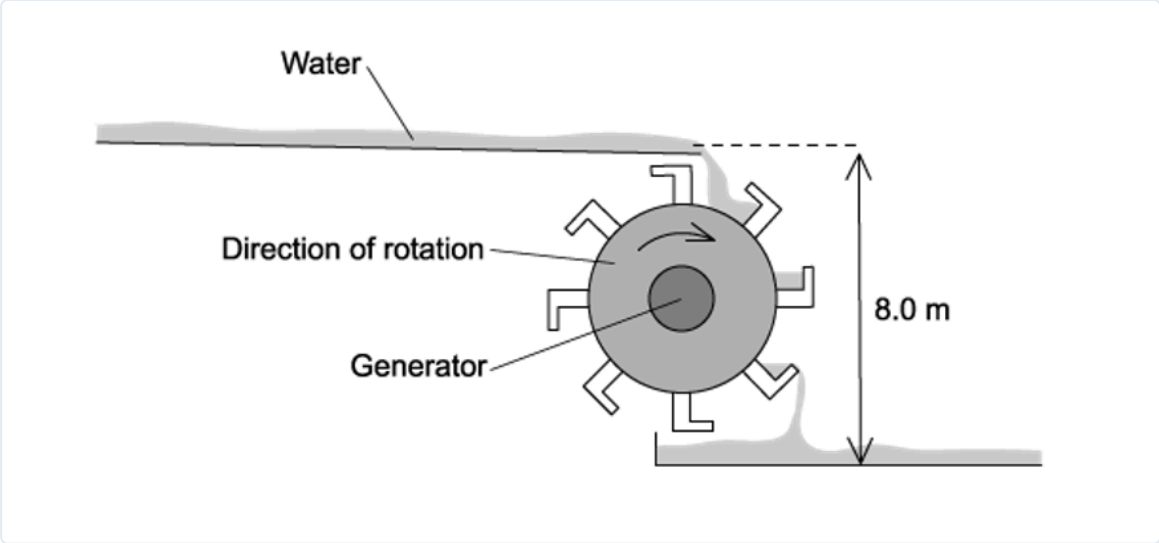
B

C

D

Question 20 Medium [1 mark]

The diagram shows the design of a water wheel which drives a generator to produce electrical energy. The flow rate of the water is 200 kg s^{-1}. The generator supplies a current of 32 A at a voltage of 230 V.



Ignoring any changes in kinetic energy of the water, what is the efficiency of the system?

A 14 %

B 16 %

C 22 %

D 47 %

Question 21 Medium [1 mark]

A conveyor belt is driven at velocity V by a motor. Sand drops vertically on to the belt at a rate of $\mathrm{m\ \ kg\ \ s^{-1}}$.

What is the additional power needed to keep the conveyor belt moving at a steady speed when the sand starts to fall on it ?

A $\frac{1}{2}mv$

B mv

C $\frac{1}{2}mv^2$

D mv^2

Question 22 Medium [1 mark]

The force resisting the motion of a car is taken as being proportional to the square of the car's speed. The magnitude of the force at a speed of $20\ \ \mathrm{m\ \ s^{-1}}$ is 800 N.

What effective power is required from the car's engine to maintain a steady speed of $40\ \ \mathrm{m\ \ s^{-1}}$?

A 32 kW

B 64 kW

C 128 kW

D 512 kW

Question 23 Medium [1 mark]

A mass is raised vertically. In a time t, the increase in its gravitational potential energy is E_p and the increase i its kinetic energy is E_k .

what is the average power input to the mass?

A $E_p - E_k t$

B $E_p + E_K t$

C $\frac{E_p - E_k}{t}$

D $\frac{E_p + E_k}{t}$

Question 24 Hard [1 mark]

A turbine at a hydroelectric power station is situated 30 m below the level of the surface of a large lake. The water passes through the turbine at a rate of 340 m^3 per minute.

The overall efficiency of the turbine and generator system is 90 %.

What is the output power of the power station? (The density of water is 1000 kg m^{-3})

A 0.15 MW

B 1.5 MW

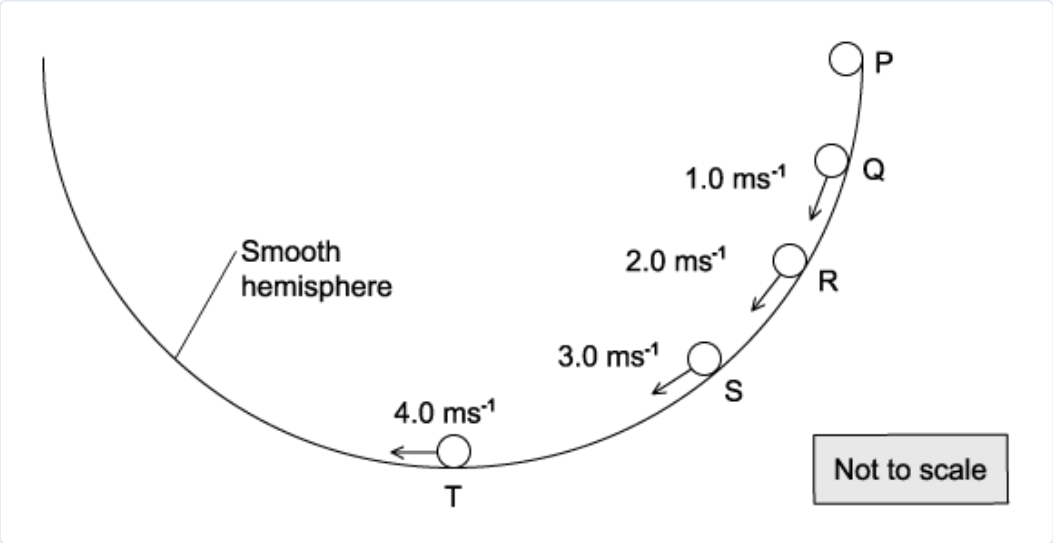
C 1.7 MW

D 90 MW

Question 25 Hard [1 mark]

A small mass is placed at point P on the inside surface of a smooth hemisphere. It is then released from rest. When it reaches the lowest point T, its speed is 4.0 m s^{-1}.

The diagram (not to scale) shows the speed of the mass at other points Q, R and S as it slides down. Air resistance is negligible.



The mass loses potential energy *E* in falling from P to T.

At which point has the mass lost potential energy $\frac{E}{4}$?

A Q

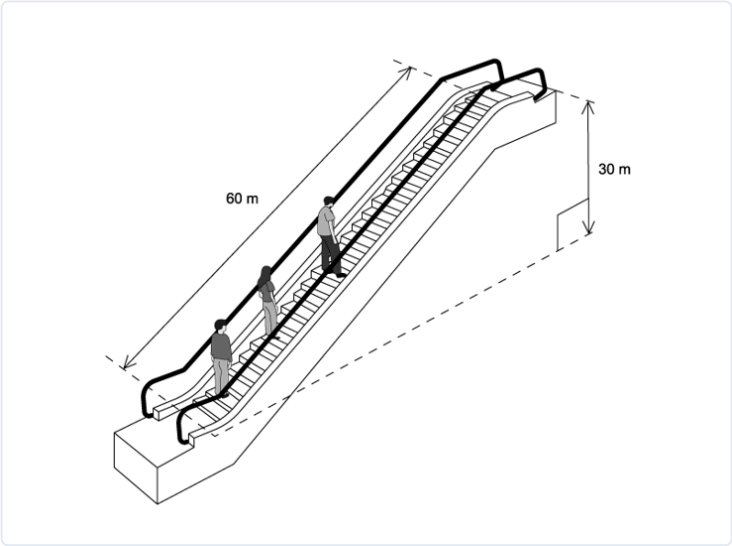
B R

C S

D none of these

Question 26 Hard [1 mark]

An escalator is 60 m long and lifts passengers through a vertical height of 30 m, as shown.



To drive the escalator against the forces of friction when there are no passengers requires a power of 2.0 kW.

The escalator is used by passengers of average mass 60 kg and the power to overcome friction remains constant.

How much power is required to drive the escalator when it is carrying 20 passengers and is travelling at 0.75 m s^{-1}?

A 4.4 kW

B 6.4 kW

C 8.8 kW

D 10.8 kW

Question 27 Hard [1 mark]

A wind turbine has blades that sweep an area of 2000 m^2. It converts the power available in the wind to electrical power with an efficiency of 50 %.

What is the electrical power generated if the wind speed is 10 m s^{-1}? (The density of air is 1.3 kg m^{-3})

A 130 kW

B 650 kW

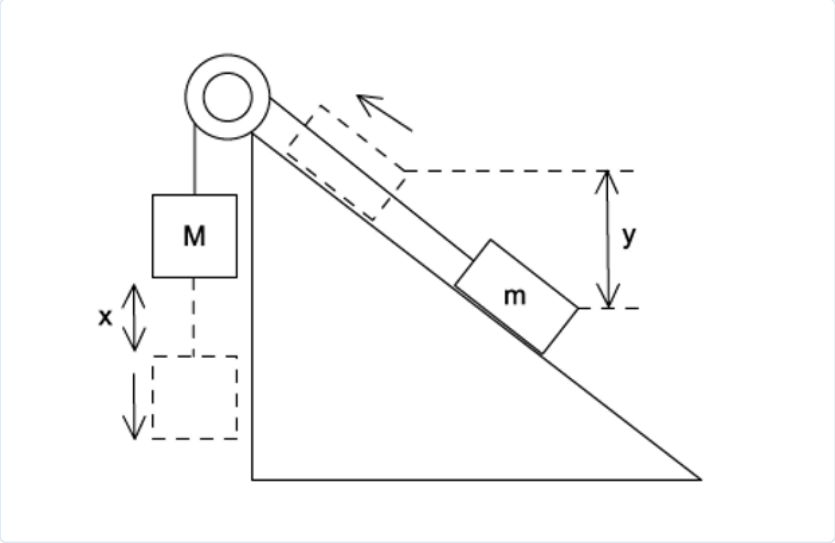
C 1300 kW

D 2600 kW

Question 28 Hard [1 mark]

The figure shows a mass m on an inclined plane connected by an inextensible string over a smooth pulley to a larger mass *M*. When the mass *M* falls through a distance *x*, the mass *m* moves up a vertical distance *y* at a constant speed.

The friction between *m* and the inclined plane is *F*.



What amount of energy is converted to heat?

A $F \quad x - y$

B Fy

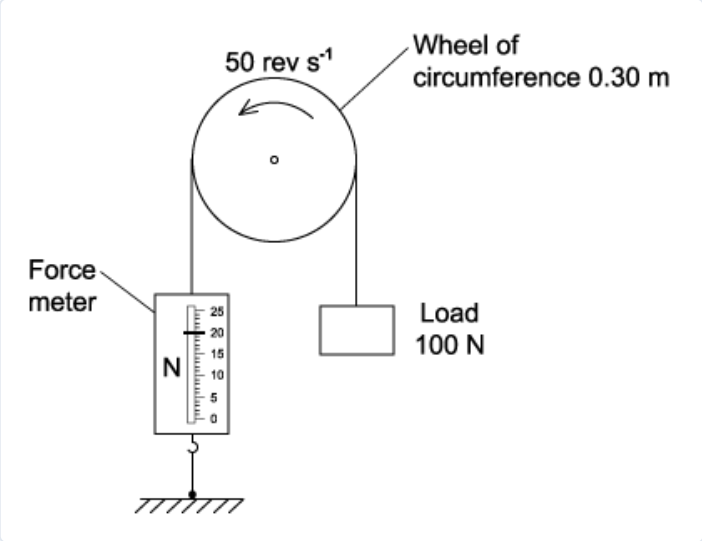
C $Mgx - mgy$

D $M - m \quad gx$

Question 29 Hard [1 mark]

The diagram shows a wheel of circumference 0.30 m. A rope is fastened at one end to a force meter. The rope passes over the wheel and supports a freely hanging load of 100 N. The wheel is driven by an electric motor at a constant rate of 50 revolutions per second.

When the wheel is turning at this rate, the force meter reads 20 N.



What is the output power of the motor?

A 0.3 kW

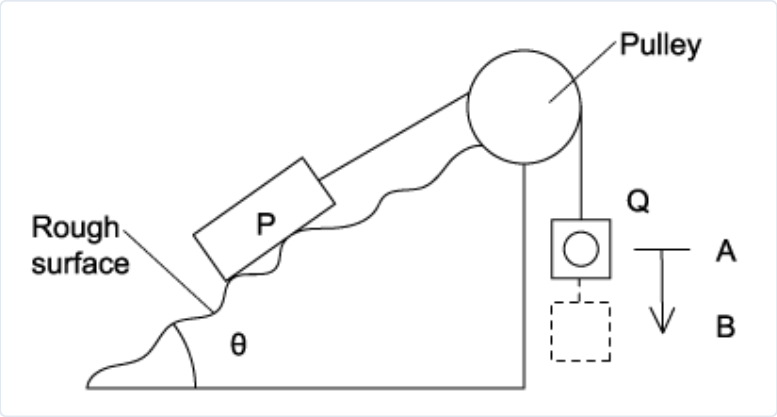
B 1.2 kW

C 1.8 kW

D 3.8 kW

Question 30 Hard [1 mark]

The figure below shows two blocks P and Q, initially stationary and connected to each other by a taut string passing over a smooth pulley. The system is released from rest and block Q moves vertically downwards from position A to B.



Which of the following statements is correct?

A Potential energy of the block Q at A = Kinetic energy of block Q at B

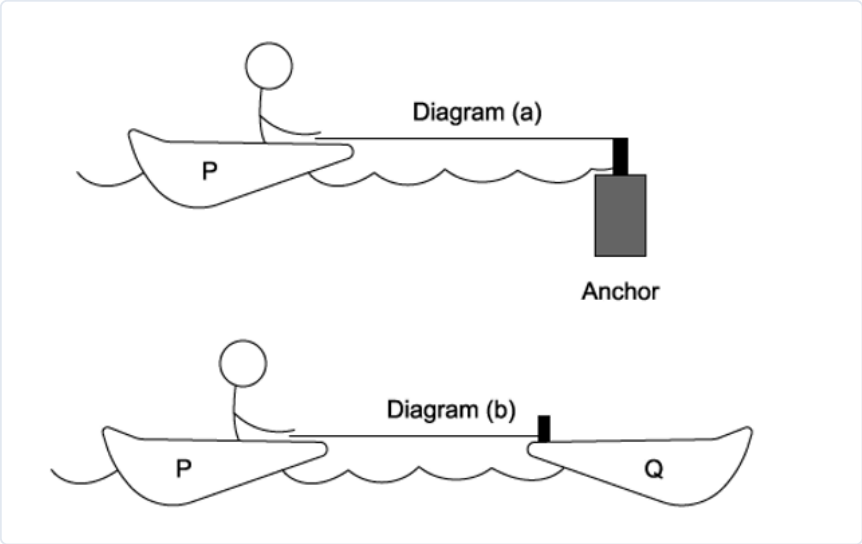
B Loss in potential energy of the whole system = Gain in kinetic energy of the whole system + work done against friction due to block P moving against a rough surface

C Loss in potential energy of the whole system = Gain in kinetic energy of the system

D Loss in potential energy of block Q = Gain in potential energy of block P

Question 31 Hard [1 mark]

A boy on a boat P pulls on a rope with a constant force *F* over a duration of time *t*. The other end of the rope is either tied to an anchor on the pier (Diagram a) or a freely floating boat Q (Diagram b) of equal mass as shown in the figures shown.



W_a and W_b represent the total work done by the boy during the time *t* and P_a and P_b represent the average power output of the boy in diagrams (a) and (b) respectively.

Which of the following statements is correct?

A $W_a > W_b \quad and \quad P_a > P_b$

B $W_a = W_b \quad and \quad P_a = P_b$

C $W_a < W_b \quad and \quad P_a < P_b$

D $W_a < W_b \quad and \quad P_a = P_b$

Question 32 Hard [1 mark]

What is the power required to give a body of mass *m* a forward acceleration *a* when it is moving with velocity *v* up a frictionless track inclined at an angle θ to the horizontal?

A $mavg \sin \theta$

B $mav \sin \theta + mgv$

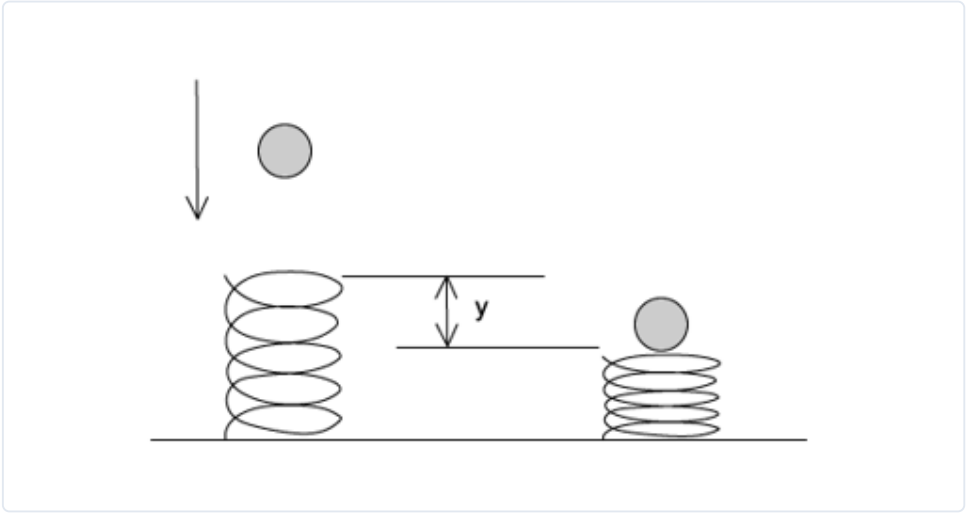
C $mav + mgv \sin \theta$

D $mav + mgv\sin \theta$

Question 33 Hard [1 mark]

A ball of mass *m* falls freely from rest. When it has reached a speed *v*, it strikes a vertical spring.

The spring is compressed by a distance *y* before the ball moves upwards again.



Assume that all energy the ball loses becomes elastic potential energy in the spring.

What is the average force exerted by the spring during its compression?

A $\frac{mv^2}{2y}$

B $\frac{m}{2y}v^2 - 2gy$

C $\frac{mv^2}{y}$

D $\frac{m}{2y}v^2 + 2gy$

Question 34 Hard [1 mark]

A hydroelectric power station has a turbine with blades that sweep through water flowing at a rate of 15 m^3 s^{-1}. The water has a density of 1000 kg m^{-3} and flows at a speed of 8.0 m s^{-1}. The turbine converts the kinetic energy of the flowing water to electrical energy with an efficiency of 40%.

What is the electrical power output of the turbine?

A 96 kW

B 190 kW

C 380 kW

D 480 kW